PROGRAMMING IN JAVA

Assignment 02

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

What is the correct signature of the `main` method in Java?

a. **public static void main(String[] args)**

b. **public void main(String args[])**

c. **void main(String[] args)**

d. **public static void main(Sting [])**

Correct Answer: a

a. **public static void main(String[] args)**

Detailed Solution:

The `main` method in Java must be declared as `public static void main(String[] args)` to be recognized by the JVM as the entry point of the program. The `public` modifier allows the method to be accessible from anywhere, `static` ensures it can be called without creating an instance of the class, and `String[] args` is the parameter used for command-line arguments.
Note: Please refer to the book Joy with Java Chapter 3 for more detailed explanation.

QUESTION 2:

Consider the code given below.

```
public class VarPrint {
    int x = 30;
    static int y = 20;

    public static void main(String[] args) {
        VarPrint t1 = new VarPrint();
        t1.x = 88;
        t1.y = 99;
        int z1 = t1.x + t1.y;
        VarPrint t2 = new VarPrint();
        System.out.println(t2.x + " " + t2.y + " " + z1);
    }
}
```

What will be the output of the above Java program?

a. **30 99 178**

b. **30 88 129**

c. **30 99 187**

d. **99 99 187**

Correct Answer: c

c. 30 99 187

Detailed Solution:

If you perform any change for instance variable these changes won't be reflected for the remaining objects. Because for every object a separate copy of instance variable will be there. But if you do any change to the static variable, that change will be reflected for all objects because a static instance maintains a single copy in memory.

Please refer to chapter 3 of book Joy With Java for a more detailed explaination.

QUESTION 3:

Consider the code given below.

```
public class ArgumenTest {
    public static void main(String[] args) {
        Test t = new Test();
        t.start();
    }

    static class Test {
        void start() {
            int a = 4;
            int b = 5;
            System.out.print("" + 8 + 3 + "");
            System.out.print(a + b);
            System.out.print(" " + a + b + "");
            System.out.print(foo() + a + b + " ");
            System.out.println(a + b + foo());


        }

        String foo() {
            return "foo";
        }
    }
}
```

What will be the output of the code given above?

a. 839 45foo45 9foo

b. 9 7 7 foo34 34foo

c. 72 34 34 foo34 34foo

d. 839 45foo45 9foo9

Correct Answer: a

a. 839 45foo45 9foo

Detailed Solution:

Here, print() methods internally converts the data in its argument into a String object and then print the composition. Here, + is the concatenation of different String representation.

Please refer to chapter 3 of book Joy With Java for a more detailed explaination.

QUESTION 4:

Which keyword is used in Java to refer to the current object?

a. **this**

b. **that**

c. **self**

d. **current**

Correct Answer: a

a. **this**

Detailed Solution:

In Java, the `this` keyword is used to refer to the current object within an instance method or a constructor. Refer to Lecture 8 for more details.

QUESTION 5:

Which of the following is true about constructors in a class?

a. **Constructors do not have a return type.**

b. **Constructors aren't used to initialize objects.**

c. **A class can have only one constructor.**

d. **Constructors cannot be overloaded.**

Correct Answer: a

a. **Constructors do not have a return type.**

Detailed Solution:

A constructor is a special method in a class that is automatically called when an object of the class is created. Its main purpose is to initialize the object's properties (variables). Unlike other methods, constructors:

- Have the same name as the class.
- Do not have a return type, not even `void`. A class can have multiple constructors with different parameter lists (constructor overloading) to allow flexibility in object creation.

Please refer book Joy with Java Chapter 3 for mored etailed explaination.

QUESTION 6:

Consider the following code snippet

```
class NPTEL_W2 {
  int x;

  NPTEL_W2(int x) {
      this.x = x;
  }

  void printX() {
      System.out.println(this.x);
  }

  public static void main(String[] args) {
      NPTEL_W2 obj = new NPTEL_W2(000);
      obj.printX();
  }
}
```

What will be the output of the code given above?

a. 0

b. 10

c. 100

d. Runtime error

Correct Answer: a

a. 0

Detailed Solution:

The constructor `NPTEL_W2 (int x)` initializes the instance variable `x` with the value passed as an argument. The method `printX()` prints the value of `x`, which is `10`. Refer to Lecture 7 for more details.

QUESTION 7:

Consider the code snippet give below.
(\n in output is to be assumed to be the new line character)

```
public class Main {
  public static void main(String[] args) {
    System.out.println("Hello");
    System.out.println("World");
    System.out.printf("Number: %d", 10);
  }
}
```

What will be the output of the code given above?

a. **Hello World\nNumber: 10**

b. **Hello WorldNumber: 10**

c. **Hello\nWorld\nNumber: 10**

d. **Hello World\nNumber: 10\n**

Correct Answer: **c**

c. **Hello\nWorld\nNumber: 10**

Detailed Solution:

The `print` method prints text without a newline, `println` prints text with a newline, and `printf` prints formatted text. The output is `Hello World` on the first line and `Number: 10` on the second line. Refer to Lecture 10 for more details.

QUESTION 8:

Which class is used in Java to take runtime data input from the user?

a. **BufferReader**

b. **UserInputStreamReader**

c. **Scanner**

d. **DataInputStreamReader**

Correct Answer: c

c. **Scanner**

Detailed Solution:

The `Scanner` class is used to take runtime data input from the user. It provides methods to read various types of input such as strings, integers, and floating-point numbers. Refer to Lecture 9 for more details.

QUESTION 9:

How do you read a line of text from the console using the Scanner class in Java?

a. **scanner.readLine()**

b. **scanner.nextLine()**

c. **scanner.getLine()**

d. **scanner.fetchLine()**

Correct Answer: b

b. **scanner.nextLine()**

Detailed Solution:

The nextLine() method of the Scanner class reads a line of text from the console. Refer to Lecture 10 for more details.

QUESTION 10:

Which of the following is the correct way to declare a class in Java?

a. **public class MyClass {}**

b. **class MyClass[] {}**

c. **public MyClass class {}**

d. **MyClass public class {}**

Correct Answer: a

a. **public class MyClass {}**

Detailed Solution:

The correct way to declare a class in Java is by using the `class` keyword followed by the class name and curly braces. Refer to Lecture 7 for more details.